Two-Type Branching Process with Pool Splitting: Analysis of Growth, Composition, and Seeding Effects

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1 Model Description

We simulate a two-type continuous-time branching process with cell types X and Y. Each cell divides independently and exponentially:

$$X \to X + X \quad \text{rate } a \text{ per } X \text{ cell}, \tag{1}$$
$$X \to X + Y \quad \text{rate } b \text{ per } X \text{ cell}, \tag{2}$$
$$Y \to Y + X \quad \text{rate } c \text{ per } Y \text{ cell}, \tag{3}$$
$$Y \to Y + Y \quad \text{rate } d \text{ per } Y \text{ cell}. \tag{4}$$

Each division event creates exactly one new cell of the appropriate type; the parent cell is not removed. All four Poisson clocks are independent. Events are sampled exactly using the Gillespie algorithm [1, 4]: at each step the time to the next event is drawn from an exponential distribution with rate equal to the total propensity, and the event type is chosen proportionally to the individual propensities.

Throughout this analysis the parameters are fixed at $a = 0.5$, $b = 0.1$, $c = 0.05$, $d = 0.4$, with initial condition $n_X(0) = 1$, $n_Y(0) = 0$. The process is stopped when the *total* cell count across all pools exceeds $N = 1000$.

1.1 Theoretical long-run composition

The mean dynamics satisfy

$$\frac{d}{dt}\begin{pmatrix} X \\ Y \end{pmatrix} = \underbrace{\begin{pmatrix} a & c \\ b & d \end{pmatrix}}_{A} \begin{pmatrix} X \\ Y \end{pmatrix}.$$

The dominant eigenvalue of A determines the exponential growth rate and the corresponding eigenvector determines the long-run composition:

$$\lambda_1 = \frac{(a+d) + \sqrt{(a-d)^2 + 4bc}}{2} = \frac{0.9 + \sqrt{0.03}}{2} \approx 0.537.$$

The stable-composition eigenvector satisfies $(A - \lambda_1 I)\mathbf{v} = 0$, giving

$$\frac{Y}{X} = \frac{\lambda_1 - a}{c} = \frac{0.037}{0.05} \approx 0.732, \qquad \frac{X}{X+Y} \approx 0.578.$$

These values are the theoretical equilibrium targets regardless of initial conditions.

1.2 Theoretical Variance of the Two Ratios

1.2.1 Moment equations for a single pool

Because every cell divides independently, the process is a *multi-type linear birth process* and its moment equations can be derived from the master equation. Define, for a pool starting from a single type-X founder:

$$\mu_X(t) = E[X(t)], \quad \mu_Y(t) = E[Y(t)],$$

$$\sigma_{XX}(t) = \mathrm{Var}(X(t)), \quad \sigma_{YY}(t) = \mathrm{Var}(Y(t)), \quad \sigma_{XY}(t) = \mathrm{Cov}(X(t), Y(t)).$$

The first moments satisfy

$$\frac{d}{dt}\begin{pmatrix}\mu_X \\ \mu_Y\end{pmatrix} = A\begin{pmatrix}\mu_X \\ \mu_Y\end{pmatrix}, \qquad A = \begin{pmatrix} a & c \\ b & d\end{pmatrix}, \tag{5}$$

so $(\mu_X(t), \mu_Y(t))^\top = e^{At}(1,0)^\top$.

Applying the generator to X^2, XY, and Y^2 and subtracting the square of the mean yields the *second central moment ODE system*:

$$\frac{d\sigma_{XX}}{dt} = 2a\,\sigma_{XX} + 2c\,\sigma_{XY} + (a\mu_X + c\mu_Y), \tag{6}$$

$$\frac{d\sigma_{XY}}{dt} = b\,\sigma_{XX} + (a+d)\,\sigma_{XY} + c\,\sigma_{YY}, \tag{7}$$

$$\frac{d\sigma_{YY}}{dt} = 2b\,\sigma_{XY} + 2d\,\sigma_{YY} + (b\mu_X + d\mu_Y), \tag{8}$$

with initial conditions $\sigma_{XX}(0) = \sigma_{XY}(0) = \sigma_{YY}(0) = 0$. This is a *linear* system driven by (μ_X, μ_Y), so it can be integrated exactly (e.g. via `scipy.integrate.odeint`).

For a pool starting with n_0 type-X founders, independence of individuals gives

$$\mathrm{Var}(X(t)) = n_0\,\sigma_{XX}(t), \quad \mathrm{Var}(Y(t)) = n_0\,\sigma_{YY}(t), \quad \mathrm{Cov}(X(t), Y(t)) = n_0\,\sigma_{XY}(t).$$

1.2.2 Delta method for the ratio variances

Let $N = X + Y$, $\alpha = \mu_X/(\mu_X + \mu_Y)$, $\beta = \mu_Y/(\mu_X + \mu_Y)$, and $r = \mu_Y/\mu_X$. Applying the delta method to the two ratios:

$$\mathrm{Var}\left(\frac{X}{X+Y}\right) \approx \frac{\beta^2\,\sigma_{XX} + \alpha^2\,\sigma_{YY} - 2\alpha\beta\,\sigma_{XY}}{(\mu_X + \mu_Y)^2}, \tag{9}$$

$$\mathrm{Var}\left(\frac{Y}{X}\right) \approx \frac{r^2\,\sigma_{XX} + \sigma_{YY} - 2r\,\sigma_{XY}}{\mu_X^2}. \tag{10}$$

Both approximations are first-order and accurate when the population is large relative to its standard deviation.

1.2.3 Long-run behavior: spectral gap and concentration

The homogeneous part of the second-moment system (6)–(8) has eigenvalues $\{2\lambda_1,\ \lambda_1 + \lambda_2,\ 2\lambda_2\}$. Because $\lambda_1 > \lambda_2$, the solution grows as $e^{2\lambda_1 t}$ for large t, so $\sigma_{XX}, \sigma_{YY}, \sigma_{XY} \sim C\,e^{2\lambda_1 t}$. At the same time $\mu_X, \mu_Y \sim C' e^{\lambda_1 t}$, and hence $(\mu_X + \mu_Y)^2 \sim e^{2\lambda_1 t}$.

The numerator and denominator of (9) thus both grow as $e^{2\lambda_1 t}$, which would naively suggest a constant limiting variance. However, the high positive correlation between X and Y (both driven by the same Malthusian martingale [2, 3] $W = \lim_{t\to\infty} e^{-\lambda_1 t}\mathbf{u}^\top \mathbf{Z}(t)$, where $\mathbf{u}$ is the left eigenvector of A normalised so that $\mathbf{u}^\top \mathbf{v}_1 = 1$) makes the numerator $\beta^2\sigma_{XX} + \alpha^2\sigma_{YY} - 2\alpha\beta\sigma_{XY}$ grow *slower* than $e^{2\lambda_1 t}$.

The correct large-t result, derived from the spectral decomposition, is

$$\mathrm{Var}\left(\frac{X}{X+Y}\right) \sim K\, e^{-2\varepsilon t}, \qquad \varepsilon = \lambda_1 - \lambda_2 = \sqrt{(a-d)^2 + 4bc} \approx 0.173, \tag{11}$$

with an analogous decay for $\mathrm{Var}(Y/X)$. The variance therefore decays *exponentially* at a rate equal to twice the **spectral gap** ε between the two eigenvalues of A.

Intuitively, the fluctuation in $X/(X+Y)$ is driven by the sub-dominant eigenvector component of $\mathbf{Z}(t)$, which decays relative to the dominant term at rate $e^{(\lambda_2-\lambda_1)t} = e^{-\varepsilon t}$. Because both X and Y share the same dominant Malthusian factor W, the W-randomness cancels in the ratio, and only the sub-dominant residual contributes to the variance.

Practical implication. Parameters with a *larger* spectral gap (e.g. increasing $a-d$ or decreasing the cross-production rates b, c) lead to faster ratio convergence and smaller variance at any given time. With our parameters, $\varepsilon \approx 0.173$, so the ratio variance halves roughly every $\ln 2/(2\varepsilon) \approx 2.0$ time units.

1.3 Implications for the split simulation

When multiple independent pools are combined, the variance of the *combined* ratio is reduced by an additional factor. If there are K_p pools each of approximate size N/K_p, and the pools are independent, then by the law of total variance the combined ratio variance is reduced by a factor of approximately $1/K_p$ relative to a single pool of the same total size — provided the seeding probability p does not introduce systematic bias. When $p \neq \alpha_{\mathrm{eq}}$ (the equilibrium fraction of X), successive pool resets introduce a systematic shift in composition, and the effective variance can actually *increase* in the early transient phase before the pools equilibrate.

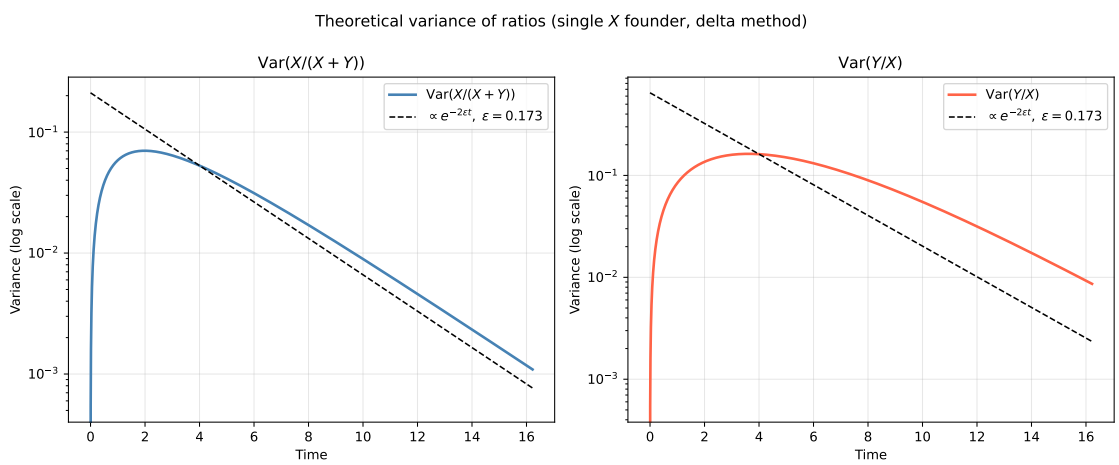

Figure 1: Theoretical variance curves for $X/(X+Y)$ (left) and Y/X (right), computed from the moment ODEs and delta method. The dashed line marks the large-t decay $\propto e^{-2\varepsilon t}$.

2 Single-Pool Baseline

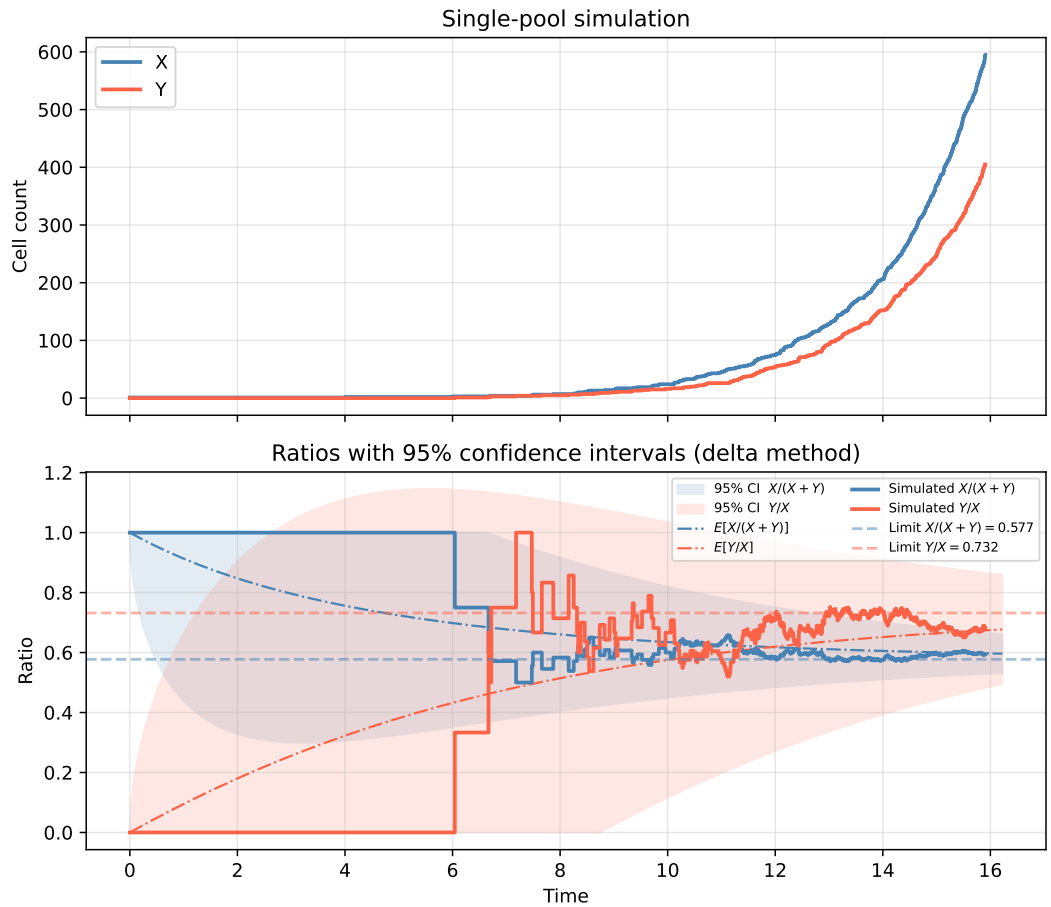

Figure 2: Single-pool Gillespie simulation [1] ($N = 1000$, $n_X(0) = 1$, $n_Y(0) = 0$). Top: counts of X and Y over time. Bottom: ratios $X/(X+Y)$ and Y/X; dashed lines show theoretical equilibrium values.

Figure 2 shows the baseline single-pool simulation. Key observations:

- **Exponential growth.** Starting from a single X cell the total population grows exponentially. The growth is initially stochastic; once the population is large the law of large numbers removes visible fluctuations.
- **X initially dominates.** Because the process starts as pure X and $a \gg b$, X cells accumulate rapidly before Y cells appear via $X \to Y$ events.
- **Convergence of ratios.** Both $X/(X+Y)$ and Y/X converge toward the theoretical equilibrium (≈ 0.578 and ≈ 0.732 respectively), illustrated by the dashed reference lines. The convergence is not monotone because of stochastic fluctuations early in the process.
- **Long-run Y/X < 1.** Y cells never exceed X cells in this regime. This follows directly from $a > d$ (X self-renews faster than Y) and the relative smallness of the cross-production rates b and c.

3 Pool-Splitting Simulation

3.1 Splitting mechanism

When the total cell count in a pool reaches the threshold K, one cell is removed and placed into a new, independent pool. The removed cell is of type X with probability p and type Y with probability $1-p$. The parent pool continues with $K-1$ cells.

In *sequential* mode (used here) pools are processed via a first-in-first-out (FIFO) queue: the active pool is simulated until it either reaches K or goes extinct, then both the parent (now $K-1$ cells) and the new child (1 cell) are placed at the end of the queue. The cumulative inter-event waiting times define a global time axis shared across all pools.

Note that a split does not change the total cell count; only division events do. Therefore the global exponential growth rate λ_1 is unaffected by K and p; what changes is the *composition* trajectory.

3.2 Effect of splitting threshold K

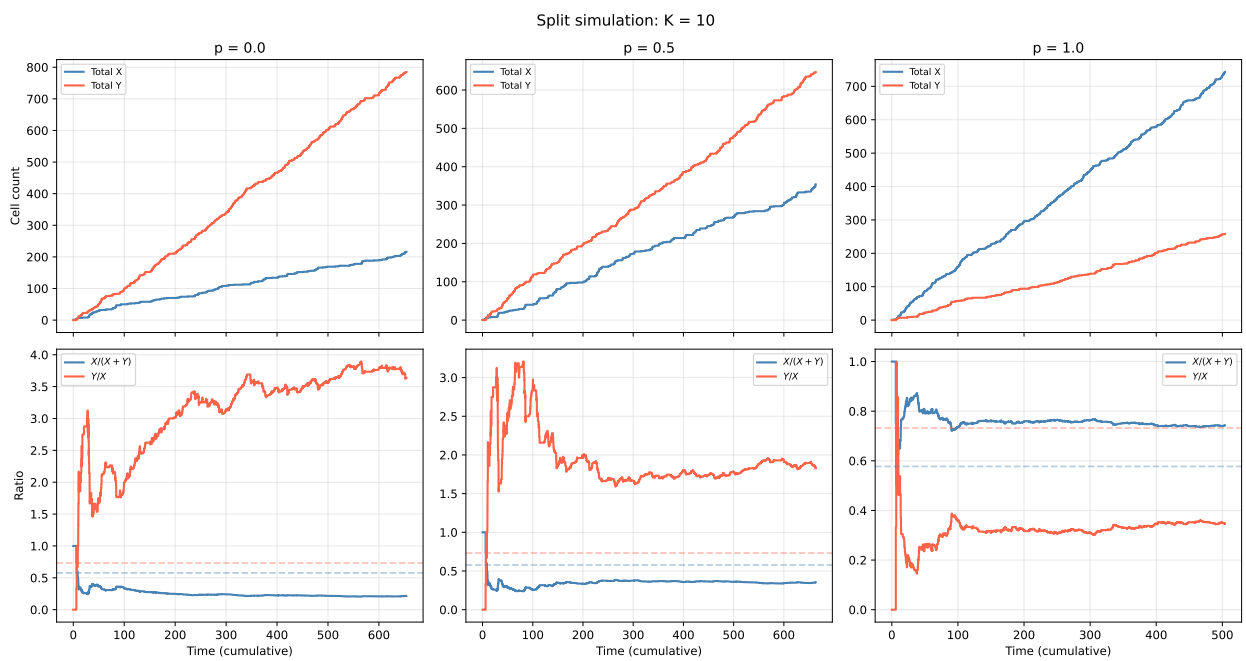

Figure 3: Split simulation with $K = 10$. Columns: $p = 0$ (left), $p = 0.5$ (centre), $p = 1$ (right). Top row: total X and Y counts. Bottom row: ratios $X/(X+Y)$ and Y/X; dashed lines mark theoretical equilibrium.

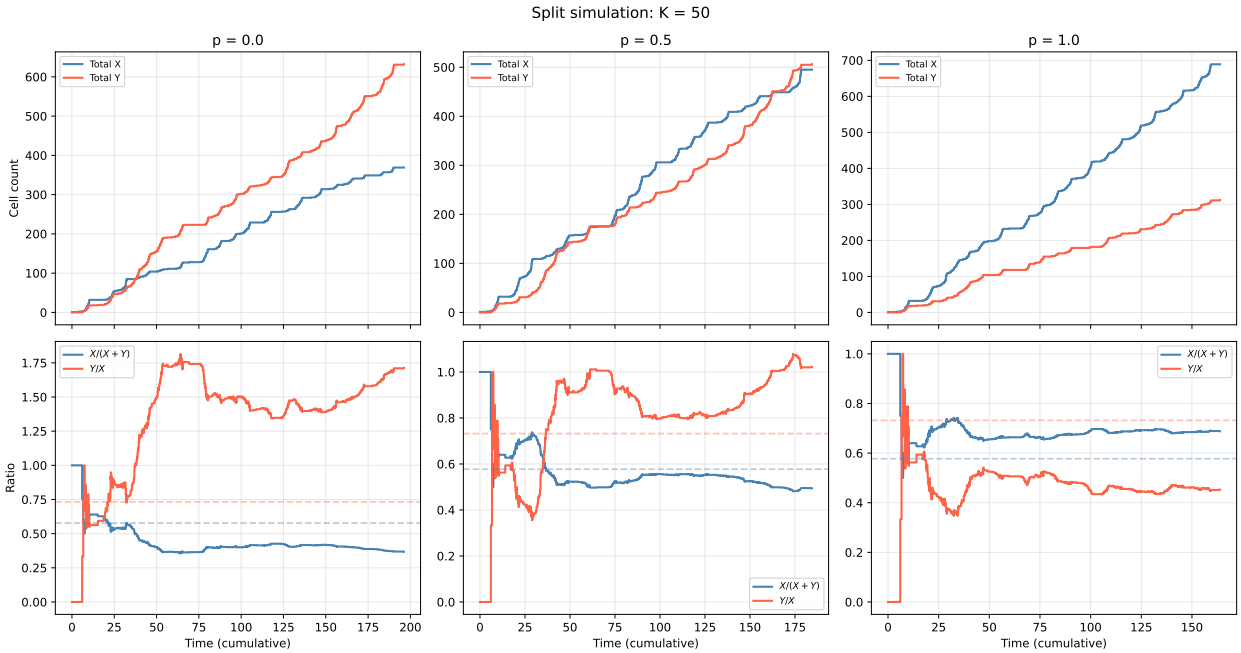

Figure 4: Same as Figure 3 for $K = 50$.

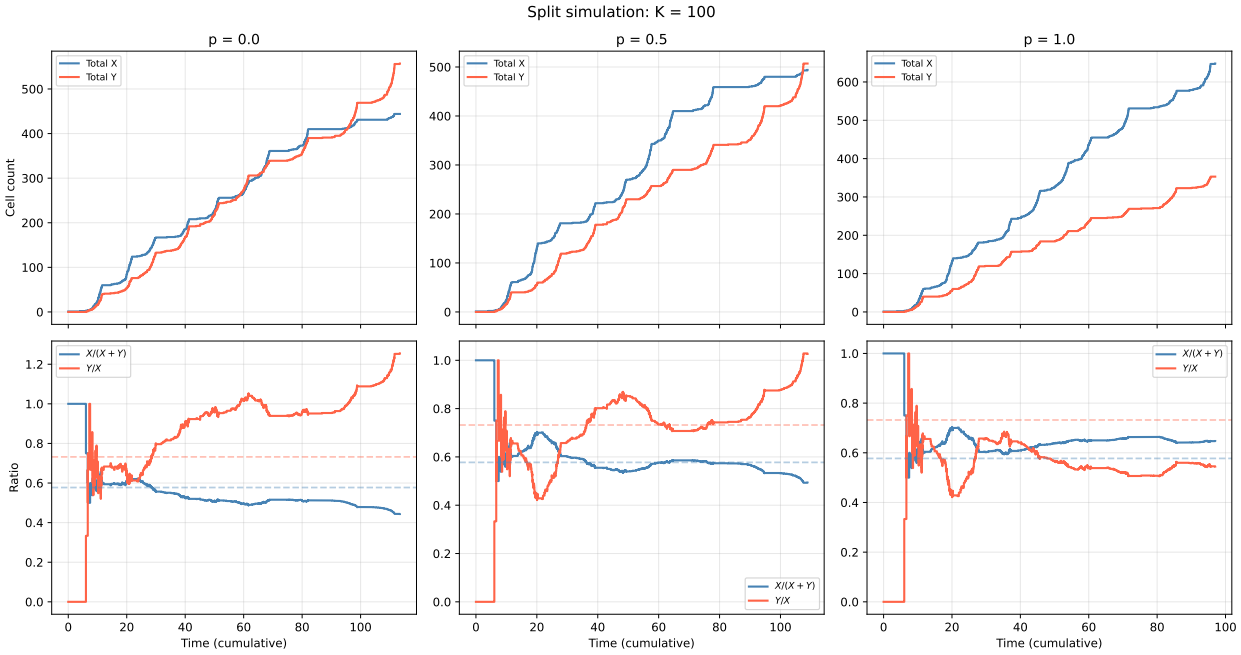

Figure 5: Same as Figure 3 for $K = 100$.

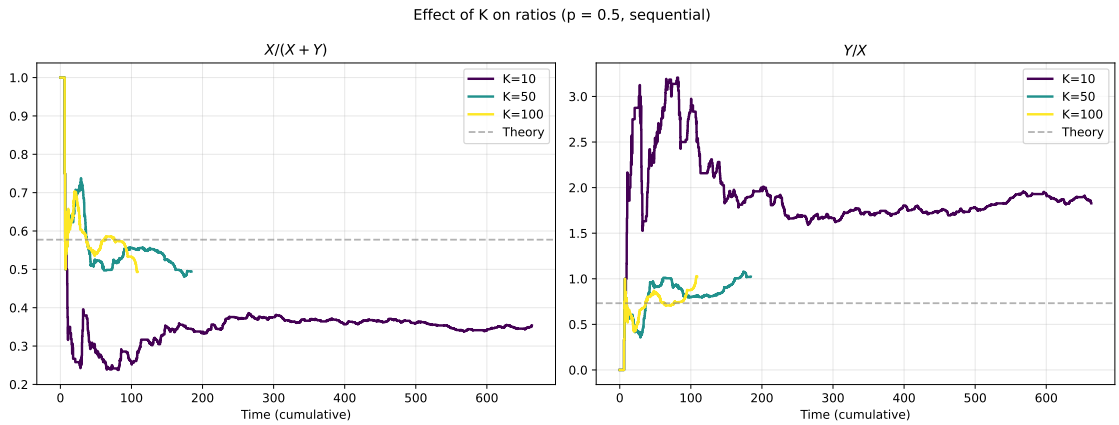

Figure 6: Cross-K comparison of ratios at $p = 0.5$. Left: $X/(X+Y)$. Right: Y/X. Dashed grey line: theoretical equilibrium.

Small K (frequent splits). With $K = 10$, each pool only undergoes $K - 1 = 9$ divisions before it splits. Starting from a single cell this is rarely enough time to approach the equilibrium composition. Consequently:

- More pools are created (lower $K \Rightarrow$ more splits for the same N), each inheriting a near-pure founder cell.
- The founder effect is strong: a pool seeded by a single X cell will be $\sim$85% X by the time it reaches $K = 10$ (since $a/(a+b) \approx 0.83$).
- The combined ratios fluctuate more and can deviate substantially from equilibrium, especially when $p \neq 0.5$.

Large K (infrequent splits). With $K = 100$, each pool runs long enough for the composition to approach the eigenvector equilibrium before it splits. As a result:

- Fewer pools are created, but each has a richer history.
- The cell sampled for transfer is drawn from a near-equilibrium pool, so it is X with probability ≈ 0.578 regardless of p (as long as p is not extreme).
- The combined ratios converge more smoothly toward 0.578 and 0.732, resembling the single-pool behaviour.

The cross-K ratio comparison (Figure 6) confirms that larger K brings the trajectories closer to equilibrium earlier in the process.

3.3 Effect of seeding probability p

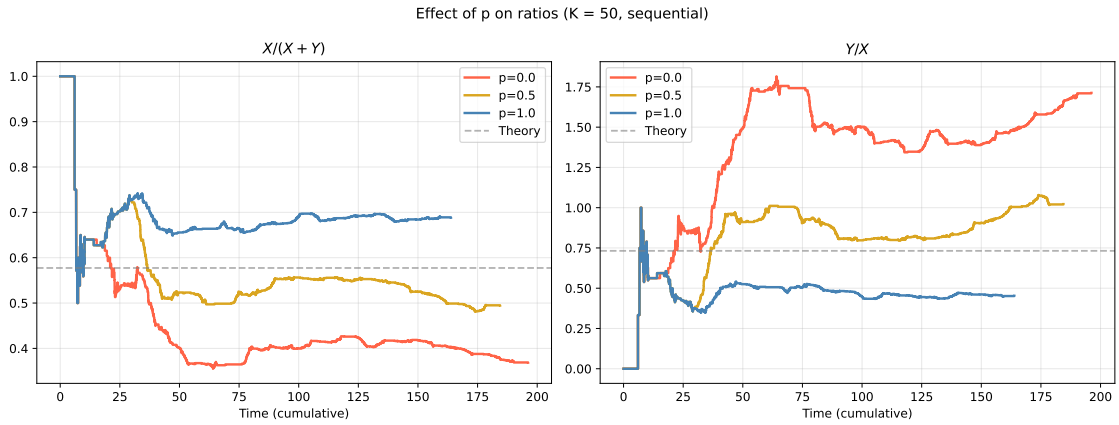

Figure 7: Cross-p comparison of ratios at $K = 50$. Left: $X/(X+Y)$. Right: Y/X. Dashed grey line: theoretical equilibrium.

The seeding probability p determines which cell type founds each new pool and therefore strongly influences the transient composition, especially for small K.

$p = 1$ **(always seed with X).** Every new pool starts as a pure X colony. Since $a \gg b$, X self-renews quickly and Y cells accumulate slowly. The combined $X/(X+Y)$ is pushed above the equilibrium value 0.578, and Y/X is correspondingly suppressed below 0.732.

$p = 0$ **(always seed with Y).** Every new pool starts as pure Y. Since $d \gg c$, Y self-renews strongly and X cells grow slowly from $Y \to X$ events. The combined $X/(X+Y)$ is pushed *below* 0.578, sometimes falling well below 0.5 for small K.

$p = 0.5$ **(equal probability).** Seeding is symmetric: roughly half of new pools are X-seeded and half are Y-seeded. The composition averages out, and the combined ratios converge toward equilibrium faster than the extreme cases.

Figure 7 shows this pattern clearly for $K = 50$: $p = 1$ keeps $X/(X+Y)$ elevated; $p = 0$ depresses it; $p = 0.5$ lies between the two and reaches the dashed equilibrium line soonest.

4 Parallel Splitting

4.1 Mechanism

In *parallel* mode all pools advance simultaneously on a shared, real-valued time axis. A global min-heap orders every pool's next potential division event by its scheduled time; at each step the soonest event fires, the affected pool is updated, and a new event is drawn from that pool. When a pool reaches K cells it immediately splits: the child pool (1 cell) enters the heap with its own first-event draw, and the parent pool (K-1 cells) continues. All pools therefore coexist and grow concurrently, unlike the sequential mode where only one pool is active at a time.

4.2 Parallel vs sequential: results

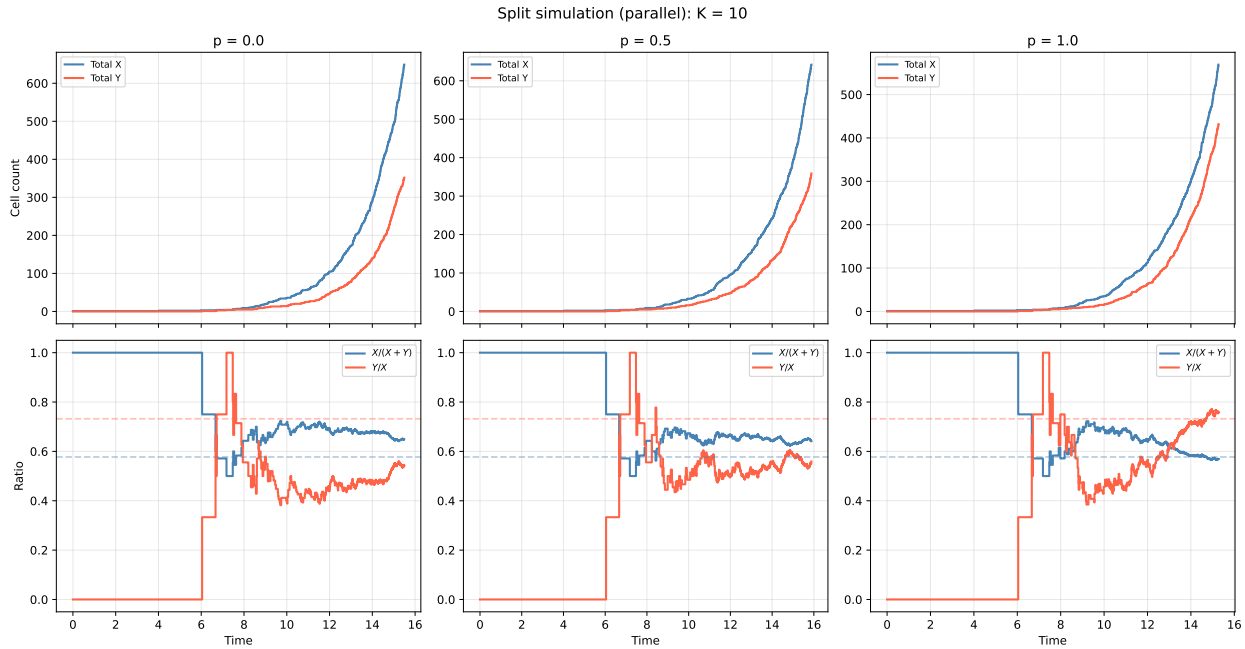

Figure 8: Parallel-mode split simulation with $K = 10$. Layout identical to Figure 3.

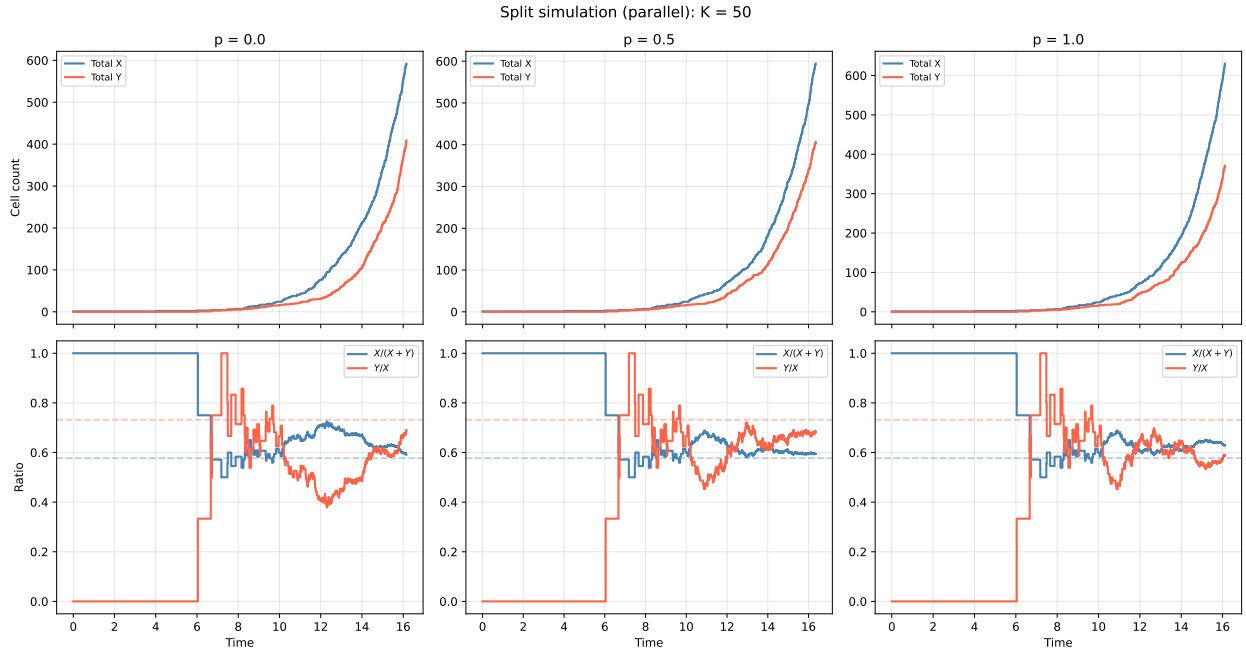

Figure 9: Parallel-mode split simulation with $K = 50$.

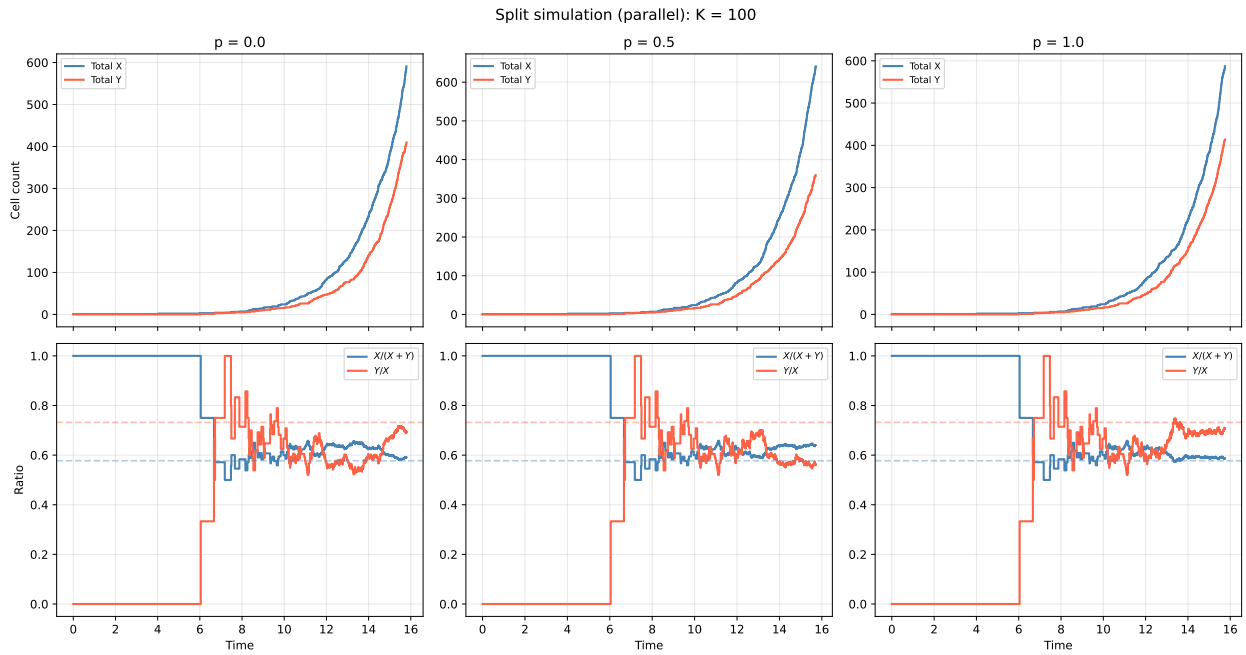

Figure 10: Parallel-mode split simulation with $K = 100$.

Comparing Figures 8–10 with their sequential counterparts (Figures 3–5) reveals a striking difference: the parallel simulations reach the theoretical equilibrium $X/(X + Y) \approx 0.578$ almost regardless of K and p, while the sequential simulations diverge markedly when $p \neq 0.5$.

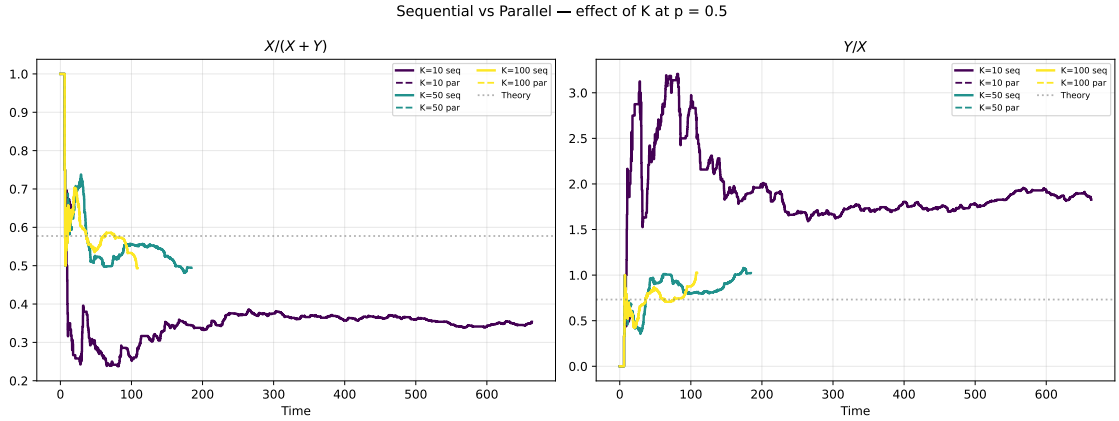

Figure 11: Sequential (solid) vs. parallel (dashed) for all three values of K at $p = 0.5$. Left: $X/(X + Y)$. Right: Y/X. Dotted grey line: theoretical equilibrium.

4.3 Why do the two modes differ so strongly?

Many simultaneous small pools in parallel. Because all pools grow concurrently, the number of active pools at any moment is large. The summary statistics show that at $K = 10$ the parallel run produces ~ 370 pools while the sequential run also produces 200 pools but visits them one at a time. Crucially, in parallel mode all pools coexist on the same time axis, so the combined ratio

is an average over many small pools *at the same time*. By the law of large numbers this average is tightly concentrated near the mean regardless of the individual founder types.

Averaging over X- and Y-founders in real time. When $p = 0$ in sequential mode, each new pool starts from a single Y cell and grows to K cells in isolation before the next pool is activated. The result is a long stretch of near-pure Y domination in the combined trajectory. In parallel mode, Y-seeded pools and residual X-rich parent pools coexist simultaneously: their X-excesses and Y-excesses cancel in the sum at every instant, leaving the combined ratio close to equilibrium.

Effect of p is suppressed. Table 2 confirms that the parallel combined composition is nearly independent of p: for $K = 10$, $X/(X + Y)$ ranges from 0.569 to 0.649 across all p, much tighter than the sequential range of 0.210 to 0.740.

4.4 Sequential vs parallel comparison figures

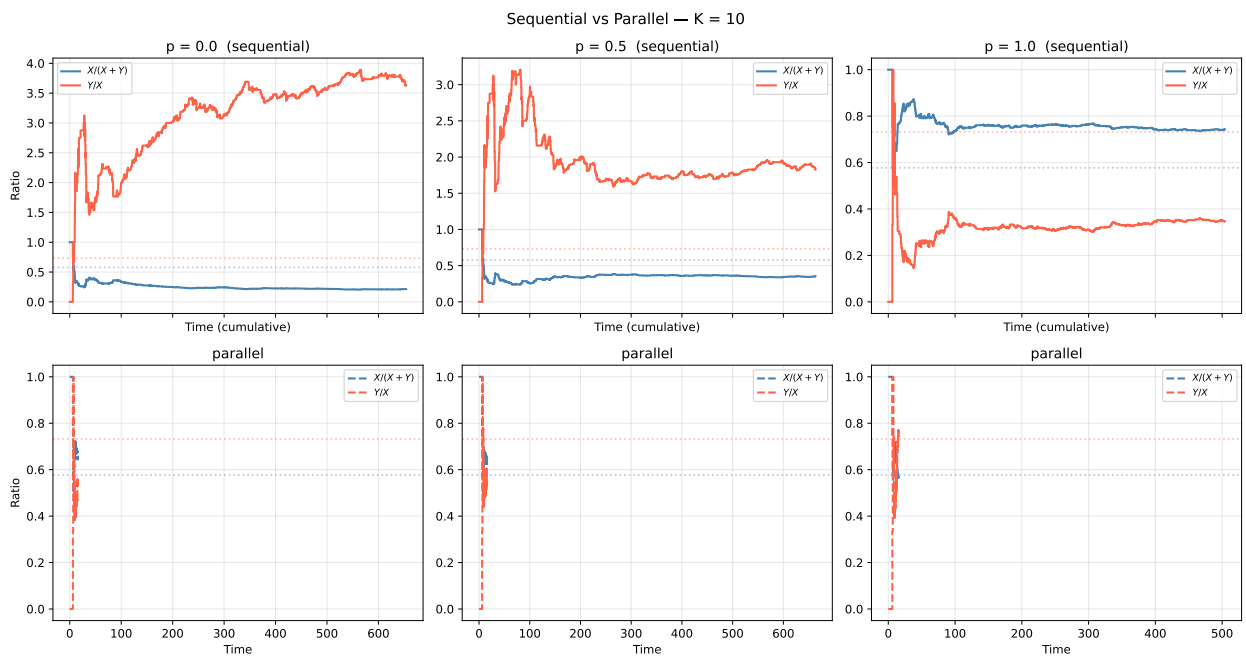

Figure 12: Direct overlay of sequential (top row) and parallel (bottom row) ratio trajectories for $K = 10$, all values of p.

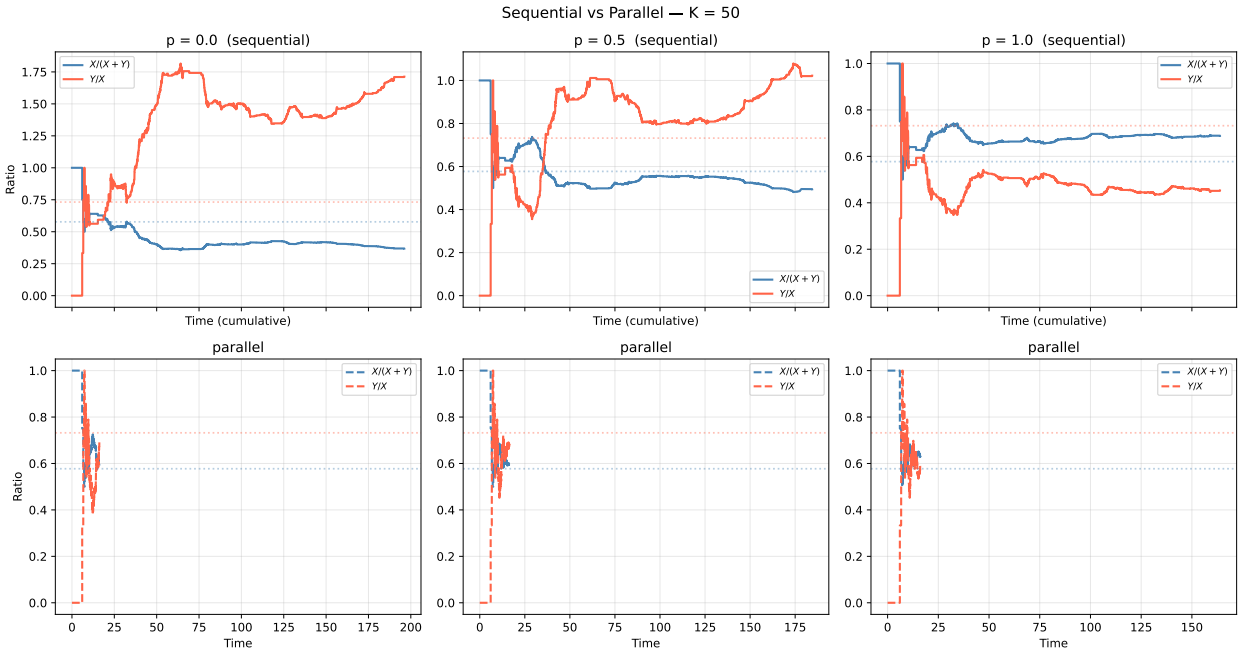

Figure 13: Same as Figure 12 for $K = 50$.

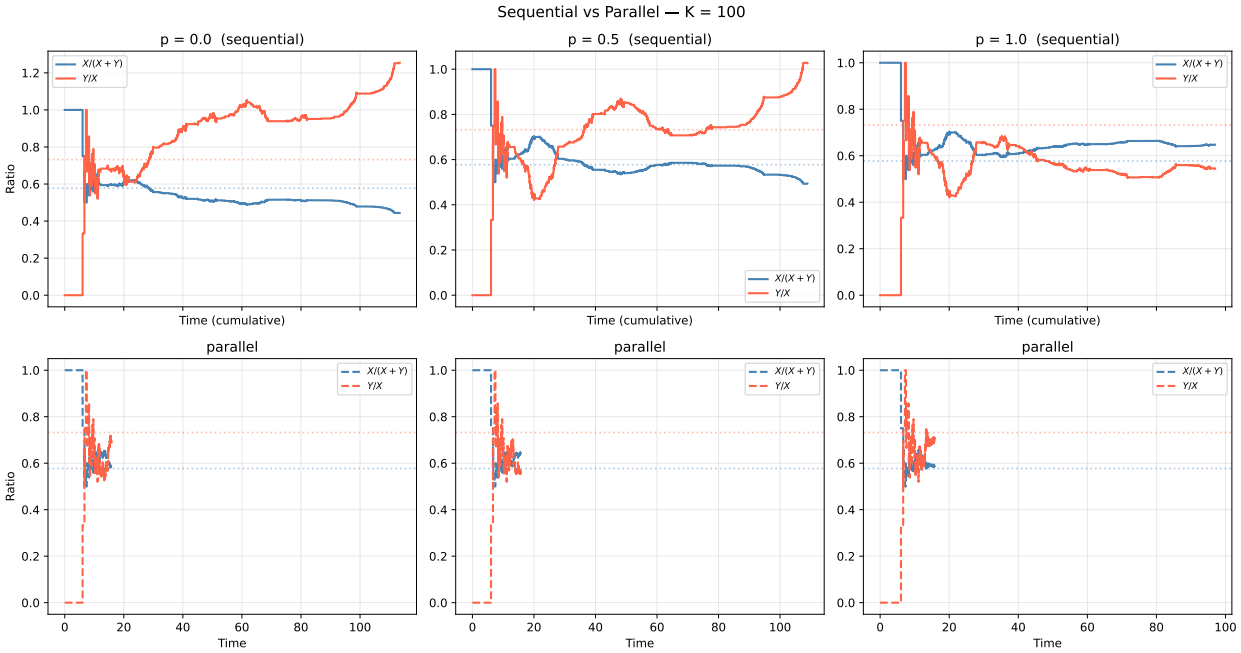

Figure 14: Same as Figure 12 for $K = 100$.

5 Comparison with the Single-Pool Scenario

Table 1 summarises the sequential results and Table 2 the parallel results.

Table 1: Sequential splitting: final composition (averages over last 10% of events). Theory: $X/(X+Y) \approx 0.578$, $Y/X \approx 0.732$.

Scenario	Pools	n_X	n_Y	$X/(X+Y)$	Y/X
Single pool	1	595	405	0.596	0.678
$K = 10$, $p = 0$	200	216	785	0.210	3.771
$K = 10$, $p = 0.5$	200	354	647	0.350	1.861
$K = 10$, $p = 1$	200	743	258	0.740	0.352
$K = 50$, $p = 0$	40	369	632	0.376	1.657
$K = 50$, $p = 0.5$	40	495	506	0.489	1.044
$K = 50$, $p = 1$	40	689	312	0.688	0.454
$K = 100$, $p = 0$	20	444	557	0.459	1.178
$K = 100$, $p = 0.5$	20	494	507	0.511	0.959
$K = 100$, $p = 1$	20	648	353	0.646	0.549
Theory (equilibrium)				0.578	0.732

Table 2: Parallel splitting: final composition (averages over last 10% of events). Theory: $X/(X+Y) \approx 0.578$, $Y/X \approx 0.732$.

Scenario	Pools	n_X	n_Y	$X/(X+Y)$	Y/X
Single pool	1	595	405	0.596	0.678
$K = 10$, $p = 0$	383	649	352	0.649	0.542
$K = 10$, $p = 0.5$	360	642	359	0.648	0.543
$K = 10$, $p = 1$	367	569	432	0.569	0.759
$K = 50$, $p = 0$	172	592	409	0.598	0.673
$K = 50$, $p = 0.5$	178	594	407	0.596	0.679
$K = 50$, $p = 1$	130	630	371	0.633	0.581
$K = 100$, $p = 0$	260	591	410	0.588	0.700
$K = 100$, $p = 0.5$	248	641	360	0.641	0.560
$K = 100$, $p = 1$	211	587	414	0.590	0.695
Theory (equilibrium)				0.578	0.732

The following points compare the sequential split, parallel split, and single-pool scenarios.

1. **Growth shape.** The single-pool simulation shows smooth near-exponential growth once the early stochastic phase passes. The sequential split simulation with large K closely mirrors this shape; small K produces a more stepped appearance because the pool queue alternates between fast-growing large pools and slow-starting single-cell pools. In parallel mode all pools grow simultaneously, so total-count growth is always smooth and exponential.

2. **Composition convergence — sequential vs. single pool.** In the single pool both types always coexist and the composition drifts gradually toward equilibrium. In the sequential split scenario with $p = 1$, new pools are repeatedly reset to pure X, maintaining an excess of

X over the equilibrium value. With $p = 0$ the excess is in Y. Only at $p = 0.5$ (or large K) does sequential behaviour closely match the single pool.

3. **Composition convergence — parallel nearly mode-independent.** The parallel mode reaches near-equilibrium composition quickly regardless of p and K (Tables 1 and 2). Concurrent growth across many pools averages out founder-type biases in real time.

4. **Variance of ratios.** Sequential splitting with small K shows substantially higher variance in the ratio trajectories compared with the single pool. Each pool reset is a bottleneck that amplifies stochastic noise (founder effect). Parallel mode suppresses this variance through averaging across pools.

5. **Y/X transient inversion.** With $p = 0$ in sequential mode, Y/X can transiently exceed 1 (more Y than X cells), something that does not occur in the single-pool scenario starting from pure X. This effect is absent in parallel mode because X-rich parent pools counterbalance Y-seeded child pools at every instant.

6 Optimal Seeding Parameters

6.1 Unbiased seeding condition

In sequential mode each new pool is seeded by a single type-X cell with probability p or a type-Y cell with probability $1 - p$. Define:

$$f_X(K) = E\left[\frac{X}{X+Y} \,\middle|\, \text{pool size} = K,\ \text{started from 1 } X \text{ cell}\right],$$
$$f_Y(K) = E\left[\frac{X}{X+Y} \,\middle|\, \text{pool size} = K,\ \text{started from 1 } Y \text{ cell}\right].$$

The expected combined X fraction is

$$\bar{\alpha}(K, p) = p\, f_X(K) + (1 - p)\, f_Y(K).$$

Setting $\bar{\alpha} = \alpha_{\text{eq}}$ and solving gives the *unbiased seeding probability*:

$$p^*(K) = \frac{\alpha_{\text{eq}} - f_Y(K)}{f_X(K) - f_Y(K)}. \tag{12}$$

Both $f_X(K)$ and $f_Y(K)$ are estimated by Monte Carlo Gillespie simulation of 2 000 independent pools per value of K.

6.2 Computed $p^*(K)$ and the bias surface

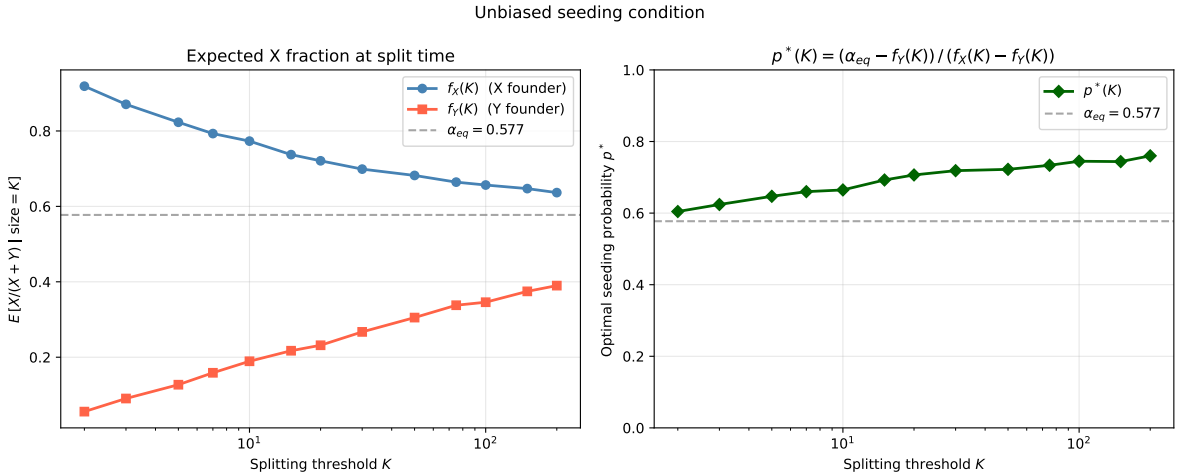

Figure 15: Left: expected X fraction at split time, $f_X(K)$ (blue) and $f_Y(K)$ (red), as a function of K. Both converge toward $\alpha_{\text{eq}} \approx 0.578$ (dashed) as $K \to \infty$. Right: the corresponding unbiased seeding probability $p^*(K)$ from (12).

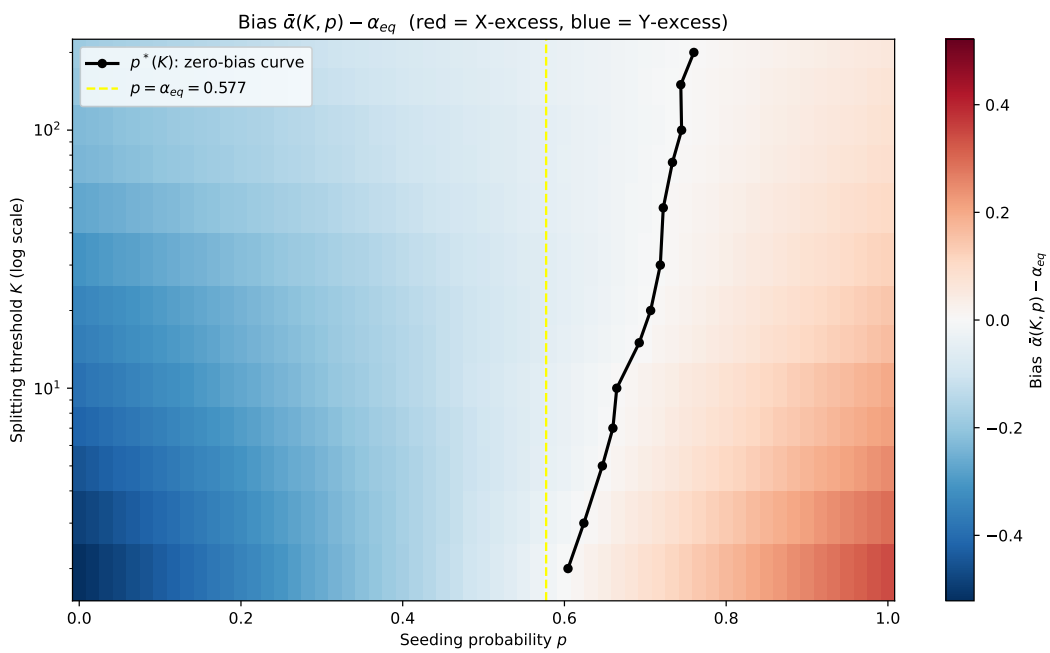

Figure 16: Bias $\bar{\alpha}(K,p) - \alpha_{\text{eq}}$ over the full (K,p) grid (log-K axis). Red indicates excess X; blue indicates excess Y. The black curve is $p^*(K)$ (zero-bias contour); the yellow dashed line is $p = \alpha_{\text{eq}}$.

Figure 15 (right) shows a key result: $p^*(K)$ does *not* converge to $\alpha_{\text{eq}} \approx 0.578$; instead it lies persistently above 0.60 for all K and is still rising toward ≈ 0.70–0.75 at $K = 200$. The reason is an *asymmetry in equilibration speed*:

- An X-seeded pool starts with self-renewal rate $a = 0.5$ and produces Y only slowly ($b = 0.1$),

so at split time it has an X-excess above α_{eq}.

- A Y-seeded pool starts with self-renewal rate $d = 0.4$ but can produce X only at rate $c = 0.05$ (much smaller than b), so its X-deficit at split time is *larger in magnitude* than the X-excess of the X-seeded pool.

Formally, using the eigen-decomposition of A (Section 1), the deviations from equilibrium at split time decay as $K^{-\varepsilon/\lambda_1} \approx K^{-0.32}$ in both cases, but with unequal prefactors. The zero-bias condition therefore converges to a nontrivial limit $p^*(\infty) > \alpha_{\text{eq}}$.

The bias surface (Figure 16) confirms that using $p = \alpha_{\text{eq}}$ (yellow dashed line) still leaves a slight X-excess for all K, while only the $p^*(K)$ curve (black) passes through zero bias.

6.3 Simulation verification at $p^*(K)$

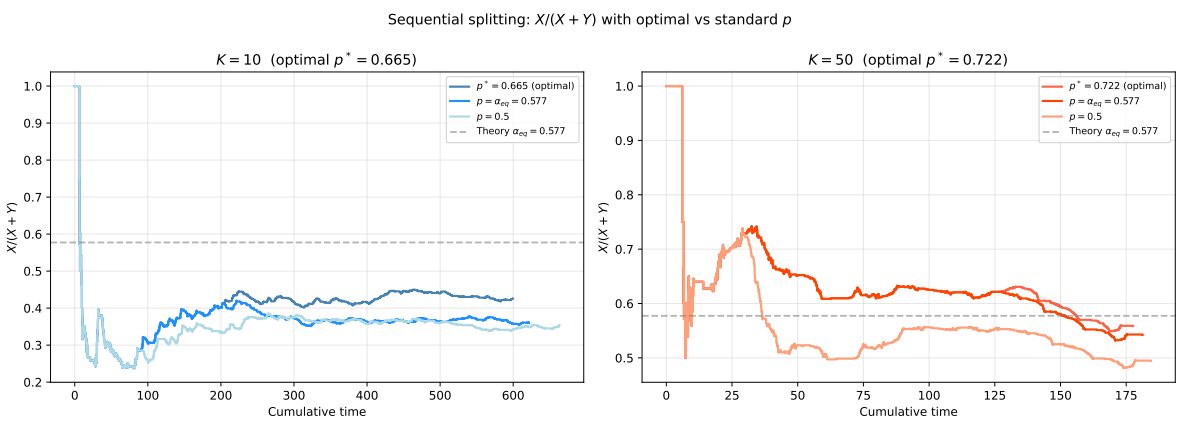

Figure 17: Sequential splitting: $X/(X+Y)$ trajectory for three choices of p at $K = 10$ (left) and $K = 50$ (right). The optimal $p^*(K)$ (dark shade) tracks the theoretical equilibrium most closely.

Figure 17 shows that seeding with $p^*(K)$ does indeed bring the combined trajectory closest to α_{eq}, while $p = 0.5$ undershoots (too many Y-seeded pools) and $p = \alpha_{\text{eq}}$ is intermediate. The improvement is especially visible at small K where the founder effect is strongest.

6.4 Variance: the K-bias trade-off

For fixed total N, the combined-ratio variance scales as

$$\operatorname{Var}\left(\frac{X}{X+Y}\right) \approx \frac{K}{N} \operatorname{Var}\left(\frac{X}{X+Y} \,\middle|\, \text{size} = K\right) \propto \frac{K^{1-2\varepsilon/\lambda_1}}{N} \propto \frac{K^{0.354}}{N}.$$

Because the exponent $1 - 2\varepsilon/\lambda_1 \approx 0.354 > 0$, the variance is *monotonically increasing in* K for our parameter values. There is therefore no interior variance-minimising K: small K always reduces variance (more pools $\Rightarrow$ more averaging), at the cost of stronger founder-effect bias.

Practical recommendation. Use $p = p^*(K)$ from (12) to eliminate bias regardless of K. Given the monotone variance result, prefer small K if variance is the priority, and use $p^*(K) \approx 0.60$–0.66. If equilibration fidelity within each pool is the priority (e.g. to avoid transient inversion of Y/X), prefer larger K and note that the bias correction is less critical since all pools nearly equilibrate before splitting.

7 Discussion

7.1 Founder effect and bottlenecks

The dominant feature of the split simulation — cell-type domination within each pool — is a manifestation of the *founder effect*: when a new population is established from a single individual, the genetic (here: type) composition of the founder disproportionately shapes the descendant population. The strength of the effect is controlled by K:

- Small K: strong bottlenecks, strong founder effect, high between-pool heterogeneity in composition.
- Large K: weak bottlenecks, composition equilibrates within each pool before the next split, between-pool heterogeneity low.

7.2 Invariance of growth rate

Because splits only redistribute existing cells (no net creation or destruction), the dominant eigenvalue λ_1 and therefore the exponential growth rate of the total population are the same in all scenarios. What changes is how quickly the composition vector $(X, Y)/N$ approaches the Perron eigenvector $\approx (0.578, 0.422)$.

7.3 Practical interpretation

If this model represents, for example, stem-cell passage in a culture dish, the splitting threshold K is the passage size and p is the fraction of stem cells (type X) deliberately maintained during passaging. The analysis shows that:

- Passaging at a large K is more "self-correcting" — the culture naturally drifts toward equilibrium composition.
- Biasing p toward 1 (seeding with X) enriches X in subsequent passages but does not prevent Y from accumulating in the long run.
- Biasing p toward 0 can temporarily invert the composition ($Y > X$) but the system eventually returns toward equilibrium as Y-seeded pools generate X cells through $Y \to X$ events.

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